4. EMOTIONS: BEFORE / AFTER

SkillWallet | Data Analytics with Tableau Capstone

Project detail	Confirmed information
Documentation date	27 October 2026
Final submission date	28 October 2026
Project name	Electric Vehicle Market Performance & Insights
Tableau workbook	Electric Vehicle Market & Technology Analysis
Team ID	LMP-EV-2026
Project owner	Yuvraj Shrivastava
Enrollment number	EN25CS3011105
Institution	Medicaps University, Indore
Contribution	Individual project completed by Yuvraj Shrivastava

Dataset and scope

The project uses the supplied electric-vehicle market CSV training dataset as the source for Tableau analysis. The supplied test CSV was retained as a project resource only and was not connected to the Tableau workbook. The confirmed preparation activities were removal of nulls, removal of duplicate records and correction of data types.

Key dataset fields

Field group	Columns / examples	Analytical purpose
Identifiers and segments	vehicle_id, model_name, manufacturer, model_year, category	Vehicle-level identification and segmentation
Battery and technical specifications	battery_type, battery_capacity_kwh, range_km, charging_speed_kw, charging_time_hr, efficiency_wh_per_km, battery_lifespan_years	Battery chemistry, range, charging and efficiency analysis
Market and commercial KPIs	price_usd, annual_sales_units, customer_rating, market_share_pct	Sales, price and customer-rating analysis
Geography and sustainability	manufacturing_country, charging_stations_count, co2_emissions_saved_tons	Geographic and sustainability context

Data preparation activities

Activity 1: Null-value handling

Null or missing values were removed during data cleaning, as confirmed by the project owner. The exact number of rows and columns affected was not provided, so no removal counts are stated.

Activity 2: Duplicate-record handling

Duplicate records were removed during data cleaning. The exact duplicate detection key and number of removed rows were not provided and are not asserted here.

Activity 3: Data cleaning and standardization

Data types were corrected to ensure fields were interpreted appropriately for analysis. Numeric measures such as range_km, price_usd, annual_sales_units, battery_capacity_kwh and customer_rating should be stored as numeric values; identifiers and categorical fields should remain dimensions/text.

Activity 4: Transformation and validation

The cleaned training CSV was used in Tableau to build aggregated measures and visualizations. The workbook uses SUM, AVG, MEDIAN, COUNTD and Percent of Total calculations as described in the project details. No Snowflake RAW/STAGING/CURATED/MART layers or SQL transformations are claimed.

Validation and use in Tableau

• The cleaned training CSV was connected to the Tableau workbook.:

• The test CSV was not connected to the Tableau workbook.:

• The project owner confirmed removal of nulls and duplicates and correction of data types.:

• Tableau worksheets use the cleaned fields for category, manufacturer, battery, country, price, range, sales and rating analysis.:

• Exact before/after row counts, null counts and duplicate counts were not supplied and should be added only if verified from the original files.:

Evidence to attach

Attach a screenshot of the CSV headers/data, a screenshot showing the cleaning steps or cleaned dataset, and a Tableau Data Source page screenshot confirming the training CSV connection. Do not include a Snowflake query output because Snowflake was not used.